# Fifth Semester B. Tech. ( Electronics and Communication Engineering ) Examination

## ( Program Elective – I )

## SMART SENSORS

Time : 3 Hours] [Max. Marks : 60

**Instructions to Candidates :—**

(1) All questions carry marks as indicated.
(2) Illustrate your answers with a neat diagram wherever necessary.
(3) All questions are compulsory.

1. (a) A sensor is designed for $-40^{0}$C to $+90^{0}$C to give output 2·5 V to 1·2 V, Deduce the following characteristics of sensor :
   (i) Range,
   (ii) Span,
   (iii) Input Full Scale,
   (iv) Output Full Scale,
   (v) Dynamic Range. 5(CO1)

   (b) Illustrate Sensitivity of a sensor with an example. How does it differ from Resolution ? 5(CO1)

2. (a) Illustrate the construction and working of LVDT Transducer with neat diagram. 5(CO2)

   (b) Explain Thermoelectric Transducers with suitable examples. 5(CO2)

3. (a) What is strain gauge ? Explain it's working principle for the measurement of stress. 5(CO2)

(b) Elaborate following Sensor fabrication techniques :

(i) Bulk Micromachining.

(ii) Surface Micromachining. 5(CO4)

4. (a) Elaborate the following protocols from IEEE 1451 family of standards :

(i) IEEE 1451·0

(ii) IEEE 1451·1

(iii) IEEE 1451·4 5(CO3)

(b) What is IEEE 1451 TIM ? Explain the transducer device interface and a transducer network interface as per IEEE 1451. 5(CO3)

5. (a) Discuss Silicon as Sensing Material. What kind of physical effects does it exhibits ? 5(CO4)

(b) Explain type of sensor having ceramics as sensing material and exhibits piezoelectric phenomenon for measurement of stress. 5(CO4)

6. (a) What properties a Smart Sensor must posses if it is to be used in automated consumer products such as smart cars or smart toys ? 5(CO5)

(b) Give the general architecture of Smart Sensor. Explain Fiber optic Sensor and Accelerometer. 5(CO5)